

User Manual: Automated Defect Triage Tool

1. Accessing the Codebase:

The complete source code and associated output files are hosted on GitHub.

- Repository Link: https://github.com/huinansky/ISE_Taks1

2. Dataset Configuration & Data Automation:

The tool is designed for seamless execution. It automatically retrieves the standardized bug report datasets directly via raw URLs from the “ideas-labo/ise-lab-solution repository”.

Users do not need to manually download any .csv files or configure local file paths.

- **Dataset Selection:** The evaluation dataset can be easily toggled. Locate the Load and Clean Data cell near the beginning of the notebook. You can switch between the datasets by commenting out the current project = '[dataset name]' line and uncommenting your target dataset (e.g., project = 'pytorch' or project = 'tensorflow').

(Note: The automated triage framework has been exclusively tested and validated on PyTorch and TensorFlow).

3. Execution Options in Google Colab:

Option A: Full Execution (From Scratch)

- **Step 1:** Download and upload the “ISE_Taks1.ipynb” file to Google Colab.
- **Step 2:** (Option But Recommended) Select Runtime -> Change runtime type -> T4 GPU.
- **Step 3:** Navigate to the top menu and click Runtime -> Run all.

Option B: Fast-Track (Visualizations Only)

To directly reproduce the final charts using the pre-computed data provided in the GitHub results/ directory:

- **Step 1: Environment Setup (Mandatory):** Execute the very first cell of the notebook. This ensures all plotting dependencies are initialized.
- **Step 2: Download Result Data:** Download the “results” folder from the GitHub repository to your local machine. It contains three sub-directories: figures, pytorch, and tensorflow/

Note on Data Consistency: The Avg_Time_Sec in the provided CSV files reflects execution on a standard CPU. While these durations differ from the GPU-accelerated timings in the report, all performance metrics (F1, Accuracy, etc.) remain identical and fully reproducible.

- **Step 3: Upload CSV Results**

1. Click the Files (folder icon) on the left sidebar of the Colab interface.

2. Navigate into the pytorch/ and tensorflow/ sub-folders within your downloaded results/ directory.

- **To visualize a single project:** Select and upload the 6 .csv files from your chosen project folder.

- **To visualize both projects:** Select and upload all 12 .csv files from both folders.

3. Drag and drop these individual .csv files directly into the Colab root directory (the space at the same level as the sample_data folder).

- **CRITICAL:** Do not upload the folders; the script is specifically designed to read the raw .csv files from the root to generate the final comparison charts.
- **Step 4: Generate Charts:** Scroll to the bottom of the notebook and manually execute only the last two cells. This will read the uploaded .csv files and generate the performance figures immediately.

4. Expected System Behavior:

Once execution begins (Option A), the tool will sequentially:

- Fetch the configured raw dataset.
- Perform automated text preprocessing (cleaning, tokenization, stopword removal).
- Extract features using both TF-IDF and SBERT.
- Train and evaluate the models (GNB, MLP, SVM) across 30 repeated loops.
- Visualizations: The final comparison figures (Model Performance Bar Chart and Statistical Significance Boxplot) will be displayed and saved to the results/figures/ directory.

5. Troubleshooting & Display Note:

"Invalid Notebook" Error on GitHub: The provided .ipynb file intentionally retains all pre-executed outputs. The GitHub web preview cannot render these dynamic widgets and may display a red error message. This is normal. Please download the notebook or open it directly in Colab for full functionality and visibility. The visualization scripts are designed to automatically pair files by their dataset prefixes (e.g., 'pytorch_') and model suffixes. Ensure all files are in the same root directory for the comparison to work.